

Kitakyushu: from steel and smog to a decarbonisation front-runner

What a Japanese heavy-industry city can and cannot do with its own hands — and why I would like to compare it with Malmö

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Subject Municipal-level industrial transition and the financing of decarbonisation

1 Why this case

Malmö and Kitakyushu are not obvious twins, but their problem was the same. Each was a second city built on a single industry — shipbuilding in Malmö, integrated steelmaking in Kitakyushu — and each lost the certainty that industry provided. Both answered by redefining themselves as environmental cities, and both are now judged on whether that redefinition produced an economy or only a reputation.

I have studied Kitakyushu since my undergraduate work, and my interest is deliberately narrow: *not* what the city declares, but who carries the capital and who carries the risk. Japanese municipalities have weak fiscal autonomy and, since liberalisation, no municipal utility in the Nordic sense. That constraint has pushed Kitakyushu towards a specific answer — buying energy services rather than owning energy assets — which I suspect is the mirror image of how Malmö, Lolland and the Danish foundation-owned firms have solved it. This note sets out the Japanese side so that our conversation can start from a shared factual base.

2 Kitakyushu at a glance

Established	1963, by the merger of five cities of equal standing (Moji, Kokura, Wakamatsu, Yahata, Tobata)
Area	491.71 km ²
Population	c. 940,000 (Jan 2022); c. 490,000 households. A designated city in long-term decline from a 1979 peak of over 1.06 million
Principal industries	Iron and steel, chemicals, cement, ceramics, automotive
Greenhouse gas emissions	14.78 Mt CO ₂ e (2019)
Share from the industrial sector	64% — against 32% for Japan as a whole (national total 1.21 Gt CO ₂ e, 2019)
Headline target	–47% GHG by FY2030 against FY2013; net zero by 2050

That 64% is the single most important number in this note. Two-thirds of Kitakyushu's emissions sit inside a handful of private industrial installations that the city does not own, cannot regulate directly, and cannot afford to lose. Every instrument described below is shaped by that fact — the city works on the third of emissions it can reach, and tries to influence the rest through industrial policy rather than climate policy.

3 Four phases of the transition

PERIOD	PHASE	WHAT HAPPENED
1901–1963	Building the steel city	The state-owned Yawata Steel Works opens in 1901 and becomes the core of Japanese industrialisation. Five towns grow together around it and merge in 1963.
1960s–1970s	Pollution and the citizens' response	Dokai Bay is declared a "sea of death"; the sky over Yahata is described as having seven colours. Local women's associations document the damage and force the issue onto the political agenda, producing pollution-control agreements between the city and industry well ahead of national law. This is the origin of the city's environmental identity — it was imposed from below, not chosen.
1980s–2000s	Environment as an industry	The clean-up is converted into an export: the Eco-Town project (from 1997) turns the Hibikinada waterfront into a recycling industrial estate; the Science and Research Park and its agency FAIS (from the 1990s) attempt to build a knowledge economy; the city becomes a hub for environmental cooperation with Asian cities. Selected as an Eco-Model City (2008) and an SDGs Future City (2018).
2010s–	Energy as an industry	Green Energy Port Hibiki (from 2011) reorients the port towards offshore wind. Zero Carbon City declaration (Oct 2020); climate emergency declaration (Jun 2021); Green Growth Strategy (Feb 2022); selection as a national <i>Decarbonisation Leading Area</i> (Apr 2022).

The structural parallel worth testing

In both cities the environmental turn began as damage control and only later became a growth story. The interesting question is whether the growth story is real. Malmö has Västra Hamnen and a university; Kitakyushu has Eco-Town and a science park. Section 6 sets out why I am sceptical about the Japanese half of that claim.

4 What Kitakyushu is doing now

4.1 Green Growth Strategy (February 2022)

An action plan for a "virtuous circle of environment and economy", built on four pillars: (i) decarbonised power and hydrogen supply, (ii) business support for innovation, (iii) decarbonised urban development through planning and transport, (iv) export of environmental business to Asia. Stated expected effects: **¥590–680 billion** of economic impact and **c. 930,000 t** of CO₂ reduction. Targets for FY2030: renewable capacity of **1,302–1,402 MW** within the city (roughly three times the current level) and hydrogen demand of **5,700 t/year**.

4.2 Decarbonisation Leading Area (April 2022)

A national programme requiring net-zero electricity in the residential and commercial sector by FY2030, intended as a replicable model — the government's phrase is a "decarbonisation domino". Kitakyushu applied *jointly with 17 neighbouring municipalities* (18 in total) and was selected in the first cohort. Scope and figures:

- Target estate: approximately **3,600 public facilities** across the 18 municipalities, plus the cluster of recycling companies in Eco-Town.
- New renewable capacity: **37 MW**; PV at approximately **290 sites** in Kitakyushu and **130 sites** in the 17 neighbouring municipalities; **100%** renewable electricity for the target facilities.
- Cascade use of second-life PV panels and EV batteries, developed with Eco-Town firms and vehicle manufacturers, to lower the cost further.
- Funded through the national Regional Decarbonisation Transition and Renewable Energy Promotion Grant.

4.3 The "100% Renewable Kitakyushu Model" (announced February 2021)

A three-step standard roadmap that any facility can be moved along:

Step 1	100% renewable electricity, supplied from generation <i>inside the city</i> (including waste-to-energy plants) through Kitakyushu Power and other retailers.
Step 2	Self-sufficient facility: PV plus battery, installed under third-party ownership. The battery charges when power is cheap and discharges when it is expensive, absorbing surplus local renewable output and cutting the bill.
Step 3	Step 2 plus energy-efficient equipment, also third-party owned, reducing total consumption and therefore the volume of renewable electricity that must be procured.

4.4 Progress on the ground

INDICATOR	DATE	RESULT
Public facilities on 100% renewable electricity (18-municipality area)	1 July 2023	626 facilities (586 in Kitakyushu itself)
CO ₂ avoided by those facilities	1 July 2023	10,936.5 t/year
Firms certified under the city's Decarbonised Electricity Certification scheme	1 May 2023	36 companies, 48 sites c. 44,550 t CO ₂ /year
PV installed under the Leading Area programme, FY2022	FY2022	16 public facilities (350 kW) and 1 Eco-Town firm (200 kW)

Batteries installed, FY2022	FY2022	14 public facilities (230 kWh) and 1 private site (10 kWh)
Air conditioning installed, FY2022	FY2022	28 public facilities; grant spend ¥100m and ¥370m respectively

The certification scheme is worth a moment. It costs the city almost nothing: a company on a decarbonised electricity tariff gets a sticker, a logo, publication on the city's channels, and — the part that bites — eligibility for a municipal finance facility and extra points in the scoring of four separate municipal grant and business-selection schemes. Recognition is converted into access to money the city already controls.

5 The mechanism most likely to interest you: third-party ownership

Kitakyushu does not buy solar panels. It buys electricity from panels that someone else owns and that happen to sit on its roofs. The city's own slogan for this is a shift "**from ownership to use**". The structure:

- Equipment company** owns and installs the PV, battery or air-conditioning unit
- **JV / consortium** operates and manages the equipment, with IoT monitoring
- **Retailer** (Kitakyushu Power and others) holds the electricity supply contract
- **Public or private facility** pays only an electricity tariff — no capital expenditure

Three design choices make this more than a financing trick:

- **A hard price rule.** The city will only do a PPA at a facility where the resulting tariff is at or below the electricity price it already pays. Decarbonisation is not permitted to cost more. This is the discipline that has kept the programme politically durable — and also what limits how fast it can spread.
- **Speed, not just cost.** The clearest case is air conditioning for school kitchens, installed for heat-stress protection of staff. Conventional municipal procurement would have taken about 13 years at roughly 10 schools a year, at c. ¥48m/year. Under third-party ownership the rollout takes about 2 years at c. ¥44m/year, and IoT monitoring extends equipment life from 13 to 15 years. The saving is real but modest; the eleven years are the point.
- **Scale by aggregation.** A city of 940,000 cannot make a low-cost PPA model bankable on its own estate alone. Bundling 18 municipalities and roughly 3,600 buildings is what creates a portfolio large enough for private counterparties to bid on.

A worked example — Hiagari Citizens' Centre (Step 2)

12 kW of PV and one 16.4 kWh stationary battery on a building consuming about 40,000 kWh a year. Roughly 13,000 kWh a year is self-consumed, giving about **33% electricity self-sufficiency**. The building is also a designated evacuation shelter: on a full charge the battery can run the office lighting for 12 hours and charge the mobile phones of 50 evacuees for 18 hours. In Japan, resilience is frequently the argument that actually unlocks the budget, with carbon as the secondary benefit — I would be interested to know whether Malmö has an equivalent "second reason".

The wider industrial build-out

- **Offshore wind.** Kitakyushu Port was designated in September 2020 as a base port under the Port and Harbour Act — the only one in western Japan. The Hibikinada offshore wind farm is planned at approximately **220 MW** (25 turbines of 9.6 MW), developed by Hibiki Wind Energy. Since 2011 the Green Energy Port Hibiki project has been assembling a four-function cluster: import/export, turbine marshalling, O&M, and related manufacturing.
- **Hydrogen.** The Fukuoka Prefecture Hydrogen Hub Council brings together around 30 organisations, chaired by the Governor of Fukuoka, with the Mayor of Kitakyushu, Kyushu Electric Power, Saibu Gas and Nippon Steel as vice-

chairs — aiming at a commercial hydrogen supply chain centred on the Hibikinada waterfront.

- **Circular industry.** Eco-Town firms recycle PV panels at a recovery rate above 99% and return aluminium, glass, silver and copper to the market; second-life EV batteries are being brought into the same system.
- **Partnership agreements as the city's main instrument.** April 2022 with Izutsuya, Toyota Motor Kyushu, Kyushu Electric Power and Kyushu Institute of Technology; October 2022 with Daikin, Denso Kyushu and Yanekara (a University of Tokyo spin-out). The city's stated ambition is to act as a *hub* — it convenes, demonstrates and certifies, because it cannot invest at scale.

6 A critical reading — where I think the case actually stands

Everything above is what the city presents, and most of it is genuinely working. My own research concerns the longer record, and it is less flattering. Kitakyushu has run an environment-and-knowledge-led growth strategy for three decades, and the city's gross product has nonetheless grown slowly relative to other designated cities. The transition has been real as environmental policy and weak as economic policy.

I examined this through a growth-contribution analysis of Kitakyushu against Fukuoka City from 1980 to 2020, and through the record of the Science and Research Park and its agency FAIS. Two findings shape how I read the current decarbonisation programme.

6.1 The industries that were targeted are not the industries that grew

PERIOD	INDUSTRIES THE CITY AND FAIS SET OUT TO GROW
1994–1998	Imaging and video, aerospace, airport-related industry
1999–2003	Environment (recycling), information technology, welfare, housing, lifestyle and culture, biotechnology
2004–2008	Environment (recycling, eco-premium products, biomass), information technology, welfare devices, biotechnology, next-generation robotics, semiconductors
2019	Semiconductors, car electronics, information and communication (system LSI), robotics

What in fact contributed most to the city's product in the most recent period was health care and social work, real estate, and miscellaneous manufacturing. The target list was rewritten roughly every five years, largely tracking whichever industry national programmes were funding at the time; the productive structure underneath it moved towards lower-value, locally-consumed services.

6.2 Three explanations I am testing

1. **The success criterion was commercialisation, not value added.** FAIS assessed projects on technical feasibility, the prospect of getting to a product, and employment — not on revenue or contribution to the city's economy. A project that reached market counted as a success even if it never grew. National schemes attached to the same projects *did* carry revenue targets, and the robotics programme, which used them, is one of the areas that actually expanded.
2. **There was never enough private capital.** No meaningful venture capital layer formed locally. Regional banks fund company formation rather than R&D, so FAIS ended up financing research from its own budget — which is bounded by a shrinking municipal balance sheet. The comparison I use is Pittsburgh, where university-led renewal was carried by roughly a billion dollars a year of investment and where venture investors brought a commercial judgement into the decision to fund.
3. **The supporting services never agglomerated.** A technology cluster needs accountants, patent and corporate lawyers, marketing, and investors within reach. Around the Science and Research Park these are thin, and the city has substituted for them by providing the services itself. That keeps the cluster running but caps it at the scale a municipality can fund, and removes the competition that would improve the services themselves.

Why this matters for our conversation

If that diagnosis is right, the third-party-ownership model in Section 5 is the most interesting thing Kitakyushu has done in thirty years — not because of the megawatts, which are small, but because it is the first instrument that puts *private balance sheets* behind municipal decarbonisation instead of asking a declining municipal budget to carry it. The unresolved question is whether buying services rather than owning assets leaves the city with any lasting capability, or whether it has simply outsourced its energy transition. That is exactly the question I want to put to a city that took the opposite route.

Section 6 reports work in progress. The contribution analysis is not yet published and the figures behind it are still being revised; I present the structural argument here rather than the numbers, and I would rather be corrected than believed.

7 What I would like to ask you

1. The shipyard.

When Kockums wound down, who absorbed the cost — the municipal balance sheet, the state, or private developers? Kitakyushu never had a single closure event; its steel industry shrank slowly and is still there, which may have made the transition politically easier and economically slower.

2. What does the City of Malmö still own?

Kitakyushu has no municipal utility. It has a stake in a local electricity retailer and it owns buildings — that is nearly the whole lever. What is Malmö's ownership position today across district heating, water, and housing, and how much of your climate programme depends on it?

3. Municipal capital, regulation, or procurement?

Of Malmö's climate targets, roughly what share is delivered by the city spending its own money, by regulating others, and by using its purchasing power? I would like to compare the mix rather than the targets.

4. The emissions you do not control.

Two-thirds of Kitakyushu's emissions belong to private industry. Does Malmö face the same asymmetry — and if so, what has actually worked: agreements, planning conditions, procurement, or public pressure?

5. Third-party ownership under EU procurement law.

Do you use PPA or energy-service models on municipal buildings? In Japan the model's weak point is that a long service contract is hard to reconcile with annual budgeting; I would expect EU procurement rules to create a different but comparable friction.

6. Cooperating with your neighbours.

Kitakyushu bundled 18 municipalities to make its programme large enough to attract private bidders. How does Malmö achieve scale with Lund, Burlöv and Region Skåne — and does joint procurement work in practice or mainly on paper?

7. The workforce.

What became of the shipyard workers? Japanese transitions are often cushioned by redeployment inside the same corporate group, which protects individuals but slows structural change. I would like to know whether Sweden's separation of employment protection from the individual firm produced a faster or a more painful adjustment — and which you would choose again.

8. Ownership forms.

I am also visiting Lolland and its holding company LOKE, and studying Danish foundation-owned enterprises. Does Malmö have anything comparable — a vehicle that lets public purposes be pursued with private capital and a long time horizon?

8 Sources

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2. City of Kitakyushu, *Green Energy Port Hibiki* — 「グリーンエネルギーポートひびき」, project brochure. — Offshore wind cluster and base port designation.
3. City of Kitakyushu, *Next Generation Energy Kitakyushu*, energy park and municipal renewable energy map. — Installed generation in the Hibikinada district.
4. Shinryo Co., Ltd. and Recycle Tech Co., Ltd., company profiles and PV panel recycling materials, 2022. — Eco-Town circular industry.
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6. Author's own research on Kitakyushu's industrial structure and the Science and Research Park (FAIS), drawing on *Kitakyushu City Economic Accounts* (1996–2020) and FAIS programme documents. Unpublished; Section 6 only.
7. Background on the pollution period and the 1963 municipal merger is drawn from the standard literature on Kitakyushu's post-war history and is not attributable to the documents above.